TECHNICAL PRESENTATION

athenea

A relight baker for 3D Gaussian splatting from production USD assets

USD meshes and materials converted into Gaussians; per-Gaussian light transport baked with the GPU path tracer; relit at draw time and judged against the path trace of the source mesh.

jesus luque · 12 October 2026



The problem and the design requirements

A trained Gaussian cloud holds the light of its capture: it cannot be relit and has no material, rig or link to an authored asset.

The converse problem: from a photoreal USD asset, produce Gaussians that **relight**, stay **faithful to a path trace** of the mesh, **keep a rig** and **scale** from far views to close-ups.

Training from path-traced views fixes the illumination and costs hours per asset; rasterising the mesh gives up the path trace's light transport.

requirement	statement
Fidelity to the path trace	Reference = path trace of the source mesh under the same lighting; quality takes precedence over frame rate.
GPU-only arithmetic	No arithmetic on scene data on the CPU: no CPU renderer, fallback or test oracle.
Pure 3DGS at draw time	Only Gaussians are drawn; the ground's interaction becomes a Gaussian shadow catcher.
Automatic, not manual	Cell size, solid vs thin glass, closed shells, the ground: detected from the data.
Small pieces, same standard	Badges, grilles, trims and lamps judged as carefully as panels, close up and far.

The system: a baker on a GPU-only renderer

THE RELIGHT BAKER

USD stage
meshes, OpenPBR materials, lights, rigs

→

mesh2splat
one Gaussian per camera cell, edges, shells, glass

→

TX bake
GPU path tracer writes each Gaussian's transfer

→

Relight
at draw time, any sky, any light

→

Judge
against the path trace of the source mesh

WHAT SUPPORTS IT

Engine
USD/Hydra, tile rasteriser, Gaussian ray tracer, path tracer; every step on scene data is a compute kernel

Animation
skinning with the full Jacobian, zonal lobes, per-part visibility

Levels of detail
per-part merge trees, error cut, culls, pages

Application
compression and a WebGPU player on a phone (parallel)

The engine: one Gaussian layout, two renderers

INTERACTIVE DRAW

Tile rasteriser

Seven compute stages (project, depth sort, gather, emit, tile sort, ranges, blend); 16×16 tiles; deterministic order.

$$\Sigma_{2D} = (J\,W\,M)(J\,W\,M)^{T},\quad M = R\,S$$
$$\Sigma' = \Sigma_{2D} + 0.3\ \mathrm{px}^2 \cdot I$$
$$k = \sqrt{\det \Sigma / \det \Sigma'}$$
$$\alpha' = k \cdot \alpha + (1 - k)/255$$

EWA projection, Mip-Splatting-style 0.3 px² filter with the opacity paid back; footprint cut at $r^2 = \min(2 \ln(255\alpha), 9)$.

GROUND TRUTH AND BAKE INTEGRATOR

Path tracer

Starts from the same visibility buffer as raster shading; next-event estimation at every vertex; the lobe stack is sampled for the bounce.

$$w_{light} = p_l^2 / (p_l^2 + p_m^2)$$
$$p_l = P(k \mid x) \cdot p_k(\omega)$$

Power-heuristic MIS; emissive triangles as a light table. Checks: convergence exponent −0.499; closed emissive shell, 0 of 3072 pixels beyond 10^{-4}.

Every image comparison is itself a GPU kernel. Blending in linear Rec.709, raster, reference and ray tracer match the analytic over-operator at p99 0.

mesh2splat: one Gaussian per camera cell

Cell from a camera

```
h = z · (A / P) / f
```

What one pixel of a camera P pixels wide covers where the piece's box is nearest to the eye (after Mesh2GS). The Corvette is baked from its camera at P = 640, 1280 and up to 5120.

Two cards 3 and 7 units away: 930 and 169 Gaussians, ratio 5.5 against $(7/3)^2 = 5.4$.

Emission

```
s_u = ‖j_u‖·σ/N
s_n = 0.1 · min(s_u, s_v)
```

Each triangle is projected on its dominant plane; a Gaussian at every cell centre inside it. σ = 1.0 (coverage 0.911, against 0.702 at the original 0.65).

A sub-cell triangle gets one Gaussian sized by its own second moments. Count, scan, write: byte-identical runs.

Opacity as coverage

```
(πσ²/2)·α² + 2πσ²·α
= -ln(1 - F)
```

A surface's coverage F becomes the per-Gaussian α by the stack's transmittance to second order, plus 1/255 for the cut tail.

Planes of coverage 0.25 / 0.5 / 0.75 draw 0.251 / 0.501 / 0.752.

Runs on the device as an AOFX effect; layered OpenPBR captured per Gaussian (base, coat, sheen, transmission, normals).

mesh2splat: edges, normal spread, shells, glass

Hard edges held

Open, folded over 40° or non-manifold edges, found by a radix sort of edge keys. Reach across each line clamped:

$$C \leftarrow C - \kappa(Cm)(Cm)^{T},\ a = \max(d/\rho,\ 0.05r)$$

Toksvig widening

Roughness widened by the spread s of normals under the Gaussian:

$$\alpha'^2 = \alpha^2 + 2s^2$$

Closed shells

n_open < 2√(1.5 T): 102 of the Corvette's 192 meshes, whose back faces may be culled.

Glass

Open or slab (2V/A under four cells): thin wall, opacity 2R/(1+R), reflection gain 1/R; else solid.



Sill panel at 1280 px. Top left: the side of the car for context. A (top right): k = 1, ρ = 1.5, selected. B: no specular filter. C: B with ρ = 1.0. B and C show a dashed dark line on the fold.

What a Gaussian stores: the TX transfer

component	values	what it holds
direct SH, degree 3	16 (scalar)	visibility × clamped cosine: geometry alone
indirect SH, degree 3	48 (rgb)	light after a bounce, coloured by what it met
reflected field, degree 3	48 (rgb)	bounced radiance by direction, under a white sky
open directions	256 bits	16 × 16 octahedral cells, set where the ray left the scene
shape operator	3 floats	(S_uu, S_uv, S_vv) in the Gaussian's axes
material marks	5 bits	the material's roughness before the bake's widening

112

f16 values per Gaussian, the TX default

480 · 256

bytes per Gaussian, file · device (first transfer: 152 · 80)

Skinned clouds: 10 values

two zonal lobes in the Gaussian's own frame

The bake: the path tracer in transfer mode

01 · SH halves

$$T_k = (1/\pi) \int V(\omega) \cdot \max(n \cdot \omega, 0) \cdot Y_k(\omega)\, d\omega$$

Stratified, uniform over the hemisphere (density $1/2\pi$): an escaping sample is worth $2\cos\theta \cdot Y_k(\omega)$. Escape at once: direct half. Escape after a bounce: indirect half, rgb.

Unoccluded point: (0.2820, 0.3258, 0.1581) against exact (0.2821, 0.3257, 0.1577).

02 · Open cells

$$T_k^{dir} = \Sigma_c \tau_c \cdot \max(n \cdot \omega_c, 0)/\pi \cdot \Delta\Omega_c \cdot Y_k(\omega_c)$$

One any-hit ray per cell of a 16 × 16 octahedral grid over the whole sphere, from four origins over the footprint. A bit is set where at least half the light passes.

The same rays give the direct half by quadrature, with no grain.

03 · Reflected field

$$F_j \approx (1/N) \Sigma_i \Omega \cdot (\beta_i / w_i) \cdot Y_j(\omega_{1,i})$$

Throughput of paths that escape after a bounce, projected on their first direction; no extra ray.

Divided by the projected indicator of closed directions, so it gives the mean radiance of what stands there.

Corvette bakes: 64 paths per Gaussian, 3 bounces. Indirect half and field go through an à-trous filter (3 iterations); the direct half, which holds the shadows, is never filtered.

Relighting at draw time: body, polish and coat

```
B = Σ_k<16 L_k (T_k^dir + T_k^ind) + E⊙·max(n·s,0)/π·O(s) + E⊙·T^ind(s)
R_polish = U g P [ O·L̃(m, r) + (1 - O)·χ·κ·F(m; α) ] + sun
R_coat = w_c E_c [ O_c·L̃(m, r_c) + (1 - O_c)·κ·F(m; α_c) ] + sun
κ = (Σ_k T_k^ind L_k + E⊙·T^ind(s)) / (√(4π)·T_0^ind)
```

Sky

16 SH coefficients of radiance; the sun taken out as an analytic lobe; a GGX-prefiltered octahedral map of 8 levels, $\tilde{L}$.

Openness O(m, α)

Cells read at the mirror and a ring of six at $\theta_r = \max(\min(1.4 \cdot \operatorname{atan} \alpha, 1.2), 2 \cdot \pi/(2 S_c))$.

Field coupling κ

The white-sky field scaled per channel by how much more the surroundings bounce under this sky. Rank one.

polishField gate χ

Cells and field for the polish only where the material's roughness ≥ 0.3; the coat always uses them.

Body, sun share and field coupling are kept per Gaussian while sky and lights stand; a frame computes only what the eye changes.

Sharp lobes per pixel, the spread counted once

1 · Per-pixel turn

$n_x = \mathrm{normalize}(n + \mathrm{clamp}_{0.5}(S \cdot e_x))$
$m(d) = \mathrm{oct}^{-1}(a_0 + \Delta a_x \cdot d_x + \Delta a_y \cdot d_y)$

Coat under r 0.2 and polish under r 0.5 read the sky per pixel, along the stored shape operator S.

2 · Unwidened roughness

$s \approx \kappa_{max} \cdot \max(\sigma_u, \sigma_v)$
$r = (r'^4 - 2s^2)^{1/4}$, held at 0

The turn already holds the spread Toksvig added at conversion; read widened, a mirror lacquer came out matte.

3 · A tube turns, a fold widens

$t = 2s$, $w = \mathrm{smoothstep}(0.7, 1.5, t)$
$S \leftarrow (1 - w) \cdot S$, $\alpha' = \sqrt{\alpha^2 + w \cdot (t/2)^2}$

A chrome tube (t in 0.3–1) keeps its thin line; a fold (t past 1) is integrated, capped at r 0.5.

4 · The pixel's own spread

$\alpha'^2 = \alpha^2 + 2 \cdot (\|h_x\|^2 + \|h_y\|^2)/12$
$\alpha' \le 1$

The box of slopes across one pixel, doubled for the mirror. In the player: specks on folds 27 / 25 → 2 / 1.

RELIT CORVETTE AGAINST THE PATH TRACE

Doubling the bake resolution from 640 to 1280 px

17.88 dB

car crop, from 16.55 dB

23.90 dB

full frame, from 22.18 dB

Display PSNR on the 900 × 420 crop of the car and on the 1920 × 1080 frame, mesh path traced from the stage camera under the autoshop_01_4k sky; 0.25 px cut with back-face culling.



Path trace · 640 px · 1280 px

The whole car along the pipeline

bake	configuration	nodes	crop	frame
640	all leaves, early transfer	5.38 M	16.37	21.92
640	error cut 1 px, both culls	675 760	16.28	21.80
640	leaves, sharp lobes unwidened	5.38 M	16.58	22.15
640	cut + polish field	—	16.55	22.18
1280	cut 0.25 px, back culling	4 998 976	17.88	23.90
1280	tube/fold split (retained)	—	17.88	23.91

≈ 0.1 dB
cost of the level of detail, at an eighth of the leaves

0.2–0.4 dB
the shading steps of the transfer, at 640 px

+1.3/+1.7 dB
the bake's resolution, crop / frame

Where the changes act: ratios to the path trace

Region mean / path trace mean (1.00 = match)

region	before	after
door, mirror to ground < 3 m	0.78	1.02
door, mirror to ground 3–30 m	0.69	0.88
door joints	0.69	1.04
door open reflections, contrast	0.35 / 0.53	0.56 / 0.74
hood highlights, offset p50	3.2 px	2.0 px
grille holes, coarse cut	2.24	1.04
sill hairs (pixels)	987	459

relMSE against the path trace, by material

step	chrome	rough metal	paint
first transfer	0.536	0.0896	0.705
TX, cells and field everywhere	1.64	0.221	0.393
TX with polish gate	0.591	0.0835	0.341
+ sharp coat and polish	0.545	0.083	0.281

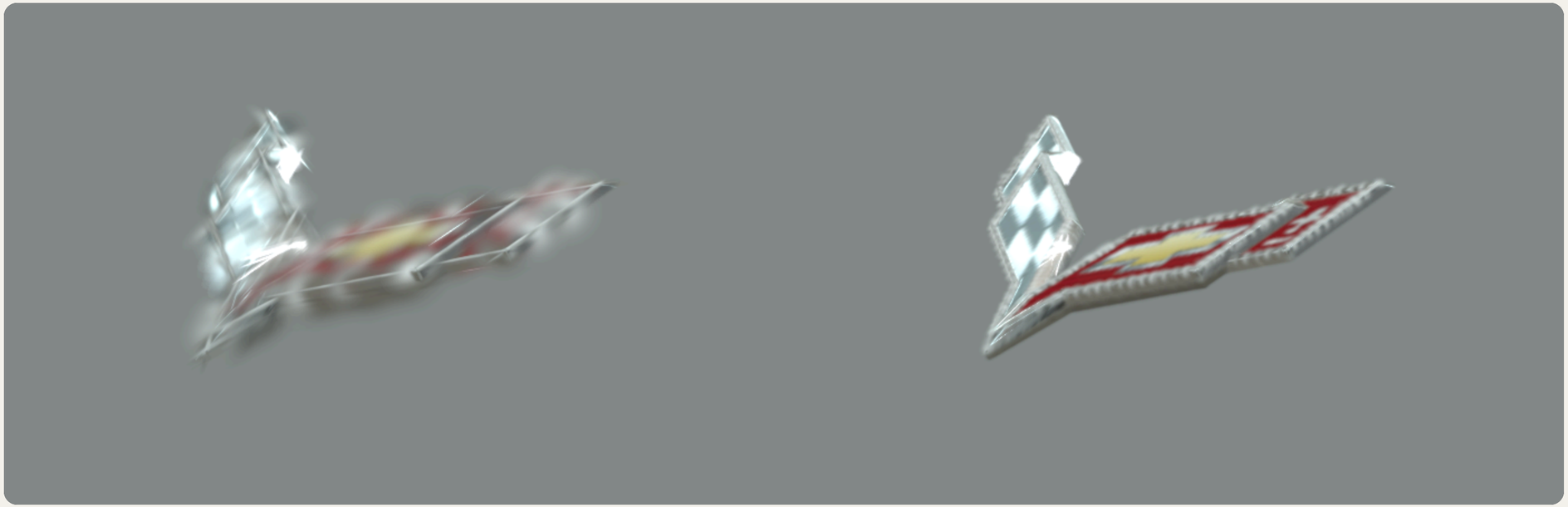
Door and mirror rows: 640 px leaves → 640 px cut with polish field. Contrast and hood offset: that cut → 1280 px cut. Grille and sill: 640 → 1280 px.



The 1280 px bake, full 1920 × 1080 frame

0.25 px cut with back-face culling, relit under the path trace's sky: 23.90 dB against the mesh path trace.

The close-up level: 5120 px



The logo from a close camera: leaves of the 1280 px bake (left) and of the 5120 px close-up level (right).

74.5 M
leaves in 24 per-part trees

148.9 M
nodes in 49 634 pages

87.4 GB
on disk

8 182
entries in one shared occlusion codebook

Animated assets: carry the rig, not the frames

$$
\begin{aligned}
\mu' &= M \, \Sigma_j \, w_j \, X_j \, (G \, \mu) \\
J &= L(M)[\Sigma_j \, w_j \, A_j \, L(G) \\
&\quad + \Sigma_j \, (X_j \, q) \, \nabla w_j^{T}] \\
\Sigma' &= J \, E \, S^2 \, E^{T} \, J^{T}
\end{aligned}
$$

Skinning, mean and covariance. Linear blend skinning of the mean; the frame pushed through the whole Jacobian, including the weights' gradients (the elastic term). Strip bent a quarter turn: 2048 of 4096 Gaussians over 3 % error → 0.

Zonal lobes. Two lobes per Gaussian, ten values, axes in its own frame; turned by the skinned frame and rebuilt into nine world SH each frame.



The sparrow: mesh path traced (left), Gaussians rasterised (right).

Per-part visibility. 12 parts; probe grids of octahedral transmittance maps baked once in the bind pose; per frame a product of table reads, no ray:

$$V = \Pi_{p \neq own} \, F_p(T_p^{-1} \, x, \, L(T_p)^{-1} \, \omega)$$

Web conversion against the mesh path trace: relMSE 0.0025.

The pawn: the static control



Path trace of the mesh (left) and Gaussians (right).

No skeleton: a marble body and a glass head on a gold ring. Every skinned path with identity transforms reduces to this rigid one.

0.0218

body relMSE, 1.57 M Gaussians, 2048^2

0.1418

glass head relMSE, same validation; three later conversions at 1024^2: 0.149, 0.081, 0.043

A solid glass is drawn as a lens; the sky through it is read at $r_t = r \cdot \sqrt{\text{sat}((\eta - 1)/\sqrt{2})}$.

Levels of detail: merge trees, a cut by error

Nodes as moments. W, Wμ, W(Σ + μμᵀ), colour, normal, roughness: a merge sums vectors, so a node holds its leaves' moments in any order. Mutual-nearest-neighbour pairs on the device; guards for part, glass, mirror, fold, highlight and colour edge.

```
ε0(n) = r_n (|cost_n| + 10·s_c(n) + θ(n)²/a_n)
ε(n) = max(ε0(n), max_k ε(k))
ρ(n) = max_k (‖μ_n - μ_k‖ + ρ(k))
P(n) = ε(n)·f / max(‖μ_n - e‖ - ρ(n), 10⁻³)
keep n ⇔ P(n) ≤ τ < P(parent)
```

One parallel test per node. P never decreases from leaf to root, so every leaf is drawn exactly once. Merge opacity by least squares: a* = Σ G·C / Σ G².

Corvette levels

level	leaves	pages
640 px	5.38 M	3 582
1280 px	8.69 M	5 789
5120 px	74.5 M	49 634

Cost against the path trace: 16.37 → 16.28 dB on the crop, at an eighth of the leaves.

Levels of detail: culls, occlusion and pages

Back faces

$\theta(n) < \pi/2 \wedge n \cdot \hat{d}_{eye} < -\sin\theta(n)$

Only on closed shells that are not glass.

Occlusion

A node's mask is the OR of its children's open cells, dilated a cell. A codebook of 8 192 entries, each a superset of its members' open sets.

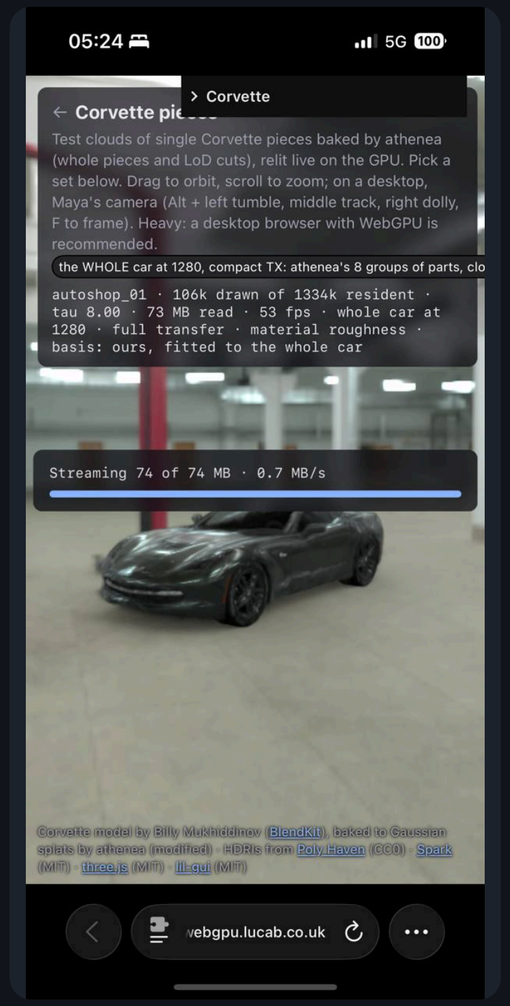
Pages by error

$b(n) = \mathrm{clamp}(\lfloor \log_2 \varepsilon(par) \rfloor + 40, 0, 63)$

A parent is never paged after its child.

cut at 393 px, against the leaves at 4×	near nodes	near dB	far nodes	far dB
error 1 px, bounds	666 585	33.29	343 541	39.15
+ back culling of closed shells	522 931	33.63	—	—
+ occlusion	360 333	33.58	195 049	39.53

Residency is the weak point: a walk needs about 2.2× the cut, and pages hold 3.6–4.5× it whatever their size.



Real iPhone, whole car, 106 k drawn, 53 fps

Application: the relit car on a phone

40-byte compact record

42.4 B resident + 16 B per drawn splat, against 308 B + 29 MiB; 40.2–41.9 dB against the full relight.

Joint CPCA on the GPU

112 values → 1 cluster id + 24 int4 weights. 512-cluster basis: 0.0181 → 0.0136 transfer relMSE on a fine group.

Open cells by VQ

256 bits → a u16 index into a shared dictionary; 62.9–65.4 dB.

Still frame

τ floor from GPU frame time (33 ms); at rest τ 0.5 px: 529 k drawn at τ 1.23, 95.6 MB of car data (emulated).

Real iPhone. Compact path, ~115 ms a million drawn + 4 ms: ~250 k at 30 fps. Close up, 160 k at 32 fps, logo soft from the 1280 px cloud, which led to the 5120 px level.

Limitations and future work

defect	cause	next step
lamps dark	no lamp emission in the transfer	bake the lamps' emission
hood highlights 2.0 px off	centre triangle's shape operator extrapolated	curvature fitted over the footprint
door open reflections 1.13	cells of ~11° let sky through	32 × 32 cells (+96 B)
chrome strip 38 % bright	a tube a few pixels across, one normal	open
far thin trims	missed by the error bounds	open

Validation gaps. One production asset under one sky; the 5120 px level shown only on a logo-and-grille test; the 5120 px data unmeasured on the phone.



Front-bumper chrome strip: path trace (top) and relit Gaussians (below).

References

- Kerbl et al. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering.
- Zwicker et al. (2001, 2002). EWA Splatting.
- Yu et al. (2024). Mip-Splatting.
- Huang et al. (2024). 2D Gaussian Splatting.
- Moënne-Loccoz et al. (2024). 3D Gaussian Ray Tracing.
- Kerbl et al. (2024). A Hierarchical 3D Gaussian Representation.
- Xiong et al. (2026). NanoGS, pairwise merging.
- Electronic Arts (SEED). mesh2splat.
- Mesh2GS. arXiv 2606.21898 (2026).
- Runnalls (2007); Karis et al. (2021), Nanite.
- Sloan, Kautz, Snyder (2002). Precomputed Radiance Transfer.
- Sloan et al. (2003). Clustered Principal Components for PRT.
- Sloan (2008). Stupid Spherical Harmonics Tricks.
- Ramamoorthi, Hanrahan (2001). Irradiance Environment Maps.
- Křivánek, Colbert (2008). Filtered Importance Sampling.
- Lagarde, de Rousiers (2014). Moving Frostbite to PBR.
- Toksvig (2005). Mipmapping Normal Maps.
- Veach, Guibas (1995). Optimally Combining Sampling Techniques.
- Andersson et al. (2024). OpenPBR Surface; MaterialX 1.39.
- Spark, a 3DGS renderer for the web; fork with a WebGPU backend.